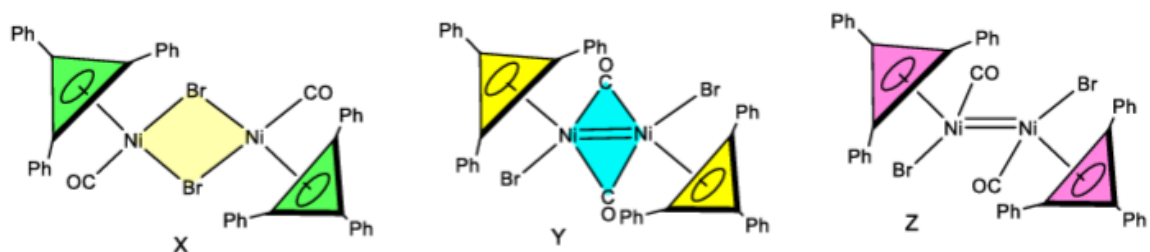


ACML 101 Inorganic Part
Home assignment -2
Organometallic Chemistry

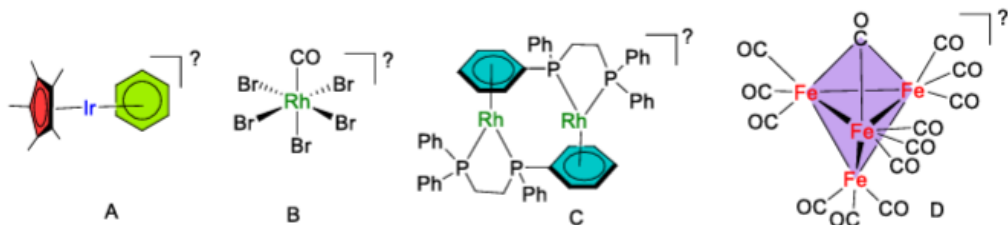
Name: Entry No. Group.....

Please answer all the questions in the provided space and submit them by **December 20, 2024**.

(1) For the dimeric complex $[(\eta\text{-C}_6\text{H}_5\text{C}_3\text{NiBr(CO)})_2]$ which one among the given three possibilities is the most likely structure? (The IR spectra of this compound gave a peak at 2039 cm^{-1}). Hints: It is generally observed that especially with first row transition metals, bridging ligands are favoured over metal-metal multiple bond formation for gaining 18 electrons.



(2) The given cationic / anionic complexes obey the 18 electron rule. The complexes having 2^+ overall charge are... (Show the electron counts)



(3) $[\text{Cp}^*\text{RuCl}]_4$ adopts a symmetrical structure having no metal–metal bonds where all the Ru centres have 18 electrons. Predict its structure ($\text{Cp}^* = \text{C}_5\text{Me}_5$)

(4) Draw the most appropriate structure of the following molecules considering that it follows 18 electron rule

(a) $[\text{Mn}(\text{CO})_4\text{Br}]_2$ and $[\text{Mn}_2(\text{CO})_4(\mu\text{-Br})]_2$

(b) $\text{Ir}_4(\text{CO})_{12}$

(c) $[\text{HRu}\{\text{P}(\text{C}_6\text{H}_5)_3\}_3]^+$

(5) For Vaska's complex, $\text{IrCl}(\text{CO})(\text{PPh}_3)_2$ which among the given activities will decrease the ν_{CO} stretching frequency of the coordinated CO ligand from 1967 cm^{-1} to lower value(s)? Justify your answer

(i) Replacing both PPh_3 by PMe_3

(ii) Oxidative addition by MeI

(iii) Replacing Cl^- by CH_3CN (cationic complex)